Skills Summary

Integrals in Context: Average Value, Accumulation, and Motion

Use this reference while completing your worksheet or when studying.

Skill 1 — Average value of a function

Rule: the average value of f on $[a, b]$ is $f_{\text{avg}} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$. Integrating alone gives an accumulated total; the division by the interval length is what turns it into a per-unit average. Sanity check: the answer must lie between the smallest and largest values of f on the interval.

Example: Find the average value of $f(x) = x^2$ on $[0, 3]$.
Step 1. The question asks for a per-unit average rather than a total, so integrate and then divide by the length of the interval.
Step 2. $\dfrac{1}{3}\displaystyle\int_0^3 x^2\,dx = \dfrac{1}{3}\left[\dfrac{x^3}{3}\right]_0^3 = \dfrac{9}{3}$.
$\Rightarrow$ $\boldsymbol{f_{\text{avg}} = 3}$

Try it: Find the average value of $f(x) = x^2$ on $[0, 6]$.

check: $\boldsymbol{f_{\text{avg}} = 12}$

Skill 2 — Accumulated change from a rate

Rule: if $r(t)$ is a rate, then $\displaystyle\int_a^b r(t)\,dt$ is the net amount accumulated between $t = a$ and $t = b$ — no division. The units multiply out and tell you whether the setup is right: litres per minute times minutes gives litres. Do *not* divide by the interval here; that would answer a different question.

Example: A tank fills at $r(t) = 6t + 1$ litres per minute. How much enters over the first 4 minutes?
Step 1. The words "how much enters" ask for a total, so integrate the rate over the time interval and do not divide by anything.
Step 2. $\displaystyle\int_0^4 (6t+1)\,dt = \left[3t^2 + t\right]_0^4 = 48 + 4$.
$\Rightarrow$ **52** litres

Try it: A tank fills at $r(t) = 8t + 3$ litres per minute. How much enters over the first 3 minutes?

check: **45** litres

Skill 3 — Displacement, distance, average velocity

Three different questions from one velocity. Displacement is $\int_a^b v\,dt$; total distance is $\int_a^b |v|\,dt$; average velocity is $\frac{1}{b-a}\int_a^b v\,dt$. Position needs a starting value: $x(t) = x(a) + \int_a^t v$. To handle $|v|$, first solve $v = 0$ to find where the sign changes, then integrate over each piece and add the absolute values.

Example: $v(t) = t^2 - 3t + 2$ m/s. Find the total distance travelled on $[0, 3]$.
Step 1. "Total distance" counts backward motion as travel, so integrate the absolute value — which means first finding where the velocity changes sign.
Step 2. $v = (t-1)(t-2)$ vanishes at $t = 1$ and $t = 2$; the three pieces measure $\frac{5}{6}$, $\frac{1}{6}$ and $\frac{5}{6}$, and adding them gives the distance.
$\Rightarrow$ $\boldsymbol{\frac{11}{6} \approx 1.833}$

Try it: Same $v(t) = t^2 - 3t + 2$, now on $[0, 4]$. Find the total distance.

check: $\boldsymbol{\frac{17}{3} \approx 5.667}$